

Determining Solutions to Inequalities*Additional Practice 7A.01.08*

Name _____

Date _____

Period _____

Use this table of possible values for questions 1 and 2.

-11.5	36	$6\frac{3}{8}$	-8	-0.1212
-25	-9.5	-7.75	$6\frac{2}{7}$	-17.3

1. Which values from the table are solutions to $-3 + x/2 < -7$?

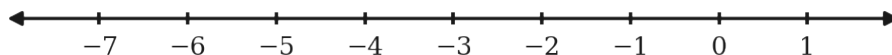
2. Which values from the table are solutions to $-3 + x/2 > -7$?

3. Rewrite both inequalities so that -8 is a solution to each. Explain your reasoning.

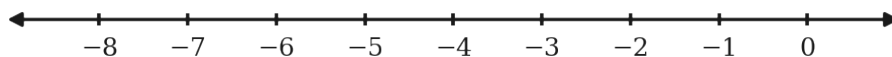
4. Complete the sentence. An inequality has _____ solutions.
5. Marco and Elena were solving $2 - 0.5x > 6$. Marco says the solution is $x > -8$. Elena says the solution is $x < -8$. Do you agree with one of them, or with neither? Justify your answer mathematically.
6. Complete the sentence. When you multiply or divide both sides of an inequality by a negative number, the inequality symbol _____.

For questions 7 through 10, solve the inequality, then graph the solution set on the number line provided.

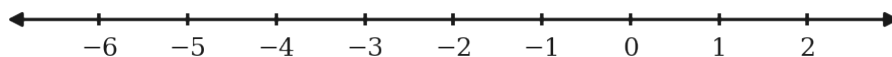
7. $-8 \leq a/(-3) - 9$



8. $9m + 16 > -20$



9. $2.8 < -1.3p - 0.5$



10. $-2 + (5/4)x \geq -12$

